

FUJITORA: 6-DoF Flight Dynamics & Powered Descent Guidance Simulator

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TECH STACK: C++17 · Python · CasADi · IPOPT · OpenGL · GLFW

REPO: github.com/OneBoyStudio/FUJITORA

FUJITORA (Flight Unit for Joint Integrated Trajectory Optimization and Recovery Analysis) is an end-to-end implementation of rigid-body 6-DoF flight dynamics coupled to a fuel-minimizing powered-descent guidance (PDG) solver. A Python/CasADi front end poses the landing problem as a constrained nonlinear program solved via direct collocation and IPOPT; a standalone C++/OpenGL engine propagates and visualizes the resulting trajectory in real time.

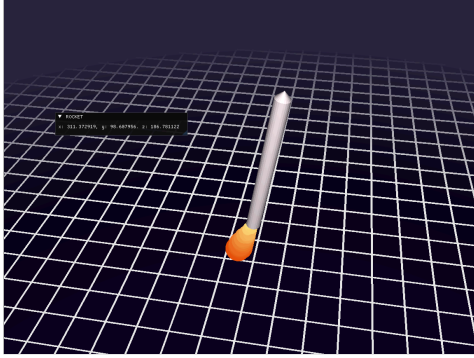


Figure 1: OpenGL/GLFW real-time render of the vehicle and launchpad during a solved descent, showing the gimbaled thrust plume.

I. SYSTEM FORMULATION

FUJITORA separates trajectory optimization from vehicle dynamics. Guidance is posed over a reduced 3-DoF point-mass model in position and velocity, $(r, v) \in \mathbb{R}^6$, with no attitude state; thrust is a free vector whose direction and magnitude are decision variables. Path and boundary constraints enforce: throttle bounded to 40 to 100% of rated thrust, a maximum tilt of 20 degrees applied to the thrust vector's angle from vertical (a proxy for vehicle attitude in this reduced model), and a 75-degree glideslope cone that keeps the descent trajectory clear of the pad until final approach, terminating at a zero-velocity touchdown. The cost functional minimizes integrated thrust magnitude as a proxy for propellant consumption.

The full-order simulation propagates a 13-state rigid body, $x = (r, v, q, \omega) \in \mathbb{R}^{13}$: inertial position r and velocity v , attitude as a unit quaternion q , and body-frame angular rate ω . No propellant slosh, structural flexibility, or aerodynamic forces are modeled; the vehicle responds only to gravity and a gimbaled thrust vector rotated into the inertial frame via q .

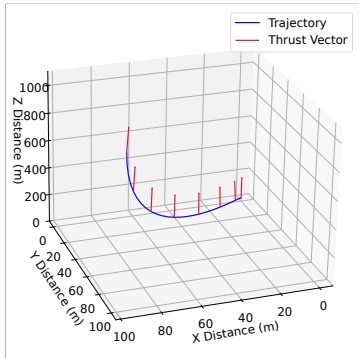


Figure 2: 3-D descent trajectory from ignition to touchdown, with instantaneous thrust vectors overlaid along the flight path.

II. NUMERICAL IMPLEMENTATION

Three layers bridge the reduced-order guidance solution to the full 13-state simulation. CasADi's symbolic framework auto-differentiates the collocation defect constraints and cost, handing exact Jacobians and Hessians to IPOPT for the point-mass trajectory problem. Its output is an open-loop reference of position, velocity, and commanded thrust vector, not a feedback policy.

An LQR controller closes the gap between that reference and the live 6-DoF state in flight. Since the guidance layer carries no attitude or angular-rate states, LQR takes the mismatch between the actual simulated state and the guidance-commanded state and produces a corrected attitude and the thrust-vector command. A separate PID controller then converts that commanded direction into the physical gimbal deflection angle that steers the rocket engine, closing the low-level actuation loop.

The full 13-state dynamics are propagated with fixed-timestep RK4 in a standalone C++ engine, which also drives an OpenGL/GLFW real-time renderer depicting the rocket body, gimbal-vector thrust plume, and launchpad geometry.

III. TELEMETRY & RESULTS

Three views characterize a solved descent: the 3-D trajectory above (Figure 2), a companion time-history panel (Figure 3) displaying velocity components, and mass depletion on a time axis (Figure 4). Together they let the throttle, tilt, and glideslope constraints be checked visually against the state trajectory rather than only against the solver's reported constraint residuals.

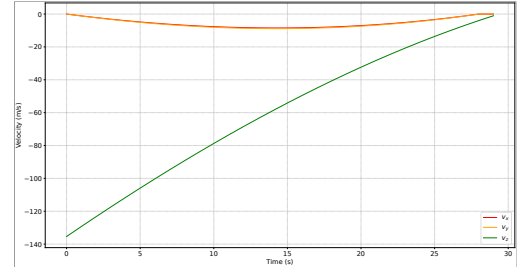


Figure 3: Velocity profiles over the descent time-history.

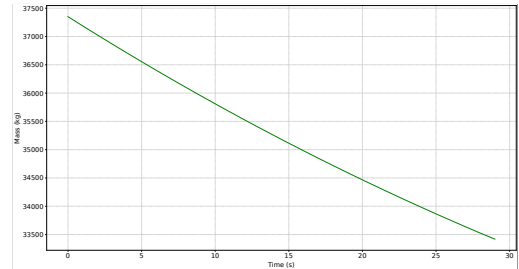


Figure 4: Mass consumption over the descent time-history.

IV. PERFORMANCE & TRADE-OFFS

The primary objective of FUJITORA is first-principles verification of the PDG pipeline end to end—from transcription and NLP solve to closed-loop dynamic execution. Trajectory tracking demonstrates robust convergence across nominal flight profiles, with solver convergence averaging under 500 ms.

The main modeling trade-off is fidelity for tractability: a rigid, aero-free vehicle keeps the guidance state space small enough for a fast, reliable IPOPT solve, at the cost of ignoring slosh and structural dynamics that matter for real vehicles. The reduced-order guidance model also pushes all attitude tracking onto the LQR/PID stack rather than solving it jointly with the trajectory; tighter coupling between guidance and control is a natural next step.